

Fractions, Decimals and Percentages — Paper B

MEDIUM • NON-CALCULATOR • 40 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Anything you cannot do in your head, do on paper.
- Give fractions in their simplest form unless the question says otherwise.
- The mark for each question is shown in square brackets on the right.

SECTION 1 – MIXED NUMBERS AND ORDER

1. Convert each number.

- (a) Write $2\frac{3}{5}$ as an improper fraction. [1]

- (b) Write $17/4$ as a mixed number. [1]

2. Work out, giving each answer as a mixed number or whole number where possible.

- (a) $1\frac{1}{2} + 2\frac{3}{4}$ [1]

- (b) $3\frac{1}{3} - 1\frac{5}{6}$ [1]

- (c) $1\frac{1}{4} \times 2\frac{2}{5}$ [2]

- 3.** Put these four numbers in order, smallest first. [2]

0.46 $\frac{4}{9}$ 43% $\frac{9}{20}$

SECTION 2 – PERCENTAGE CHANGE

- 4.** Increase 250 by 12%. [2]

- 5.** Decrease 640 by 15%. [2]

- 6.** A shirt cost 800 rupees. It is now sold for 680 rupees. Calculate the percentage decrease in price. [3]

- 7.** The population of a town rises from 4000 to 4600. Calculate the percentage increase. [3]

SECTION 3 – WORD PROBLEMS

- 8.** Work out $\frac{3}{8}$ of 320. [2]

- 9.** Which is larger: $\frac{5}{8}$ of 400, or 60% of 420? You must show your working and state which one is larger. [3]

- 10.** A car park has 250 spaces and is 68% full. How many spaces are empty? [2]

- 11.** Write 0.375 as a fraction in its simplest form and as a percentage. [2]

- 12.** In a class of 40 students, 24 are girls. What percentage of the class are boys? [3]

Two steps. Find the number of boys first — the question is not asking about girls.

End of paper. Check every answer has a unit or a form the question actually asked for.